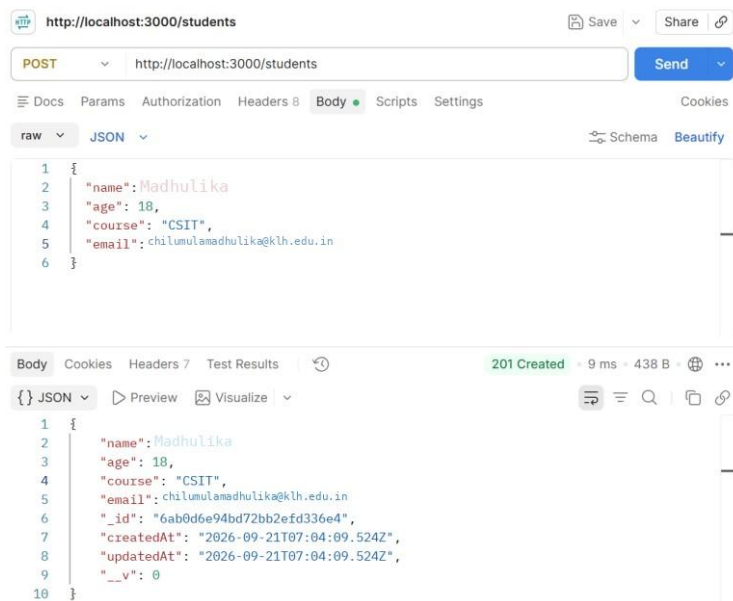


WEEK-7 SKILL — Student Records CRUD API [Niyathi Konduparthy, 2520090072]

POST:



http://localhost:3000/students

POST http://localhost:3000/students

Send

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

raw JSON Schema Beautify

```
1 {
2   "name": "Madhulika"
3   "age": 18,
4   "course": "CSIT",
5   "email": "chilumulamadhulika@klh.edu.in"
6 }
```

Body Cookies Headers 7 Test Results 201 Created · 9 ms · 438 B

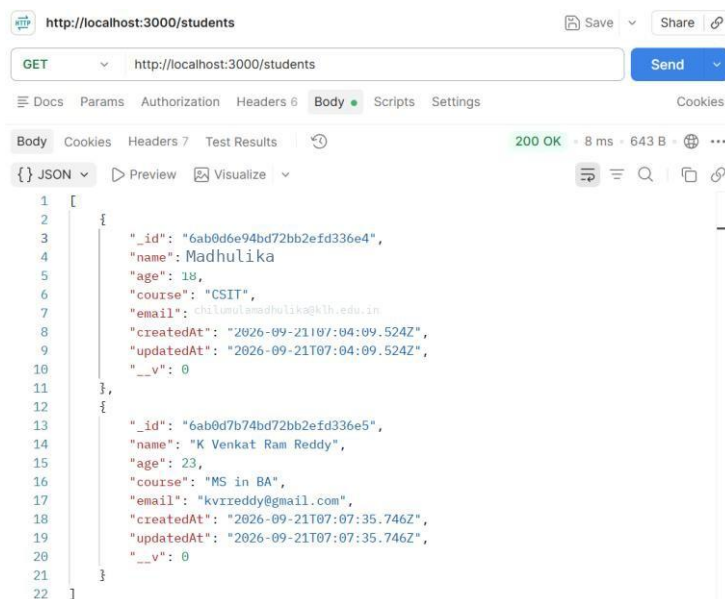
JSON Preview Visualize

```
1 {
2   "name": "Madhulika"
3   "age": 18,
4   "course": "CSIT",
5   "email": "chilumulamadhulika@klh.edu.in"
6   "_id": "6ab0d6e94bd72bb2efd336e4",
7   "createdAt": "2026-09-21T07:04:09.524Z",
8   "updatedAt": "2026-09-21T07:04:09.524Z",
9   "__v": 0
10 }
```

After POST in mongosh:

```
student_records> db.students.find()
[
  {
    _id: ObjectId('6ab0d6e94bd72bb2efd336e4'),
    name: Madhulika
    age: 18,
    course: 'CSIT',
    email: chilumulamadhulika@klh.edu.in
    createdAt: ISODate('2026-09-21T07:04:09.524Z'),
    updatedAt: ISODate('2026-09-21T07:04:09.524Z'),
    __v: 0
  }
]
```

GET all data:



http://localhost:3000/students

GET http://localhost:3000/students

Send

Docs Params Authorization Headers 6 Body Scripts Settings Cookies

Body Cookies Headers 7 Test Results 200 OK · 8 ms · 643 B

JSON Preview Visualize

```
1 [
2   {
3     "_id": "6ab0d6e94bd72bb2efd336e4",
4     "name": "Madhulika"
5     "age": 18,
6     "course": "CSIT",
7     "email": "chilumulamadhulika@klh.edu.in"
8     "createdAt": "2026-09-21T07:04:09.524Z",
9     "updatedAt": "2026-09-21T07:04:09.524Z",
10    "__v": 0
11  },
12  {
13    "_id": "6ab0d7b74bd72bb2efd336e5",
14    "name": "K Venkat Ram Reddy",
15    "age": 23,
16    "course": "MS in BA",
17    "email": "kvireddy@gmail.com",
18    "createdAt": "2026-09-21T07:07:35.746Z",
19    "updatedAt": "2026-09-21T07:07:35.746Z",
20    "__v": 0
21  }
22 ]
```

GET all data in mongosh:

```
student_records> db.students.find()
[
  {
    _id: ObjectId('6ab0d6e94bd72bb2efd336e4'),
    name: Madhulika
    age: 18,
    course: 'CSIT',
    email: chilumulamadhulika@klh.edu.in
    createdAt: ISODate('2026-09-21T07:04:09.524Z'),
    updatedAt: ISODate('2026-09-21T07:04:09.524Z'),
    __v: 0
  },
  {
    _id: ObjectId('6ab0d7b74bd72bb2efd336e5'),
    name: 'K Venkat Ram Reddy',
    age: 23,
    course: 'MS in BA',
    email: 'kvrreddy@gmail.com',
    createdAt: ISODate('2026-09-21T07:07:35.746Z'),
    updatedAt: ISODate('2026-09-21T07:07:35.746Z'),
    __v: 0
  }
]
```

GET one with specific id:

http://localhost:3000/students/6ab0d6e94bd72bb2efd336e4

GET http://localhost:3000/students/6ab0d6e94bd72bb2efd336e4

200 OK · 9 ms · 433 B

```
{
  "_id": "6ab0d6e94bd72bb2efd336e4",
  "name": "Madhulika",
  "age": 18,
  "course": "CSIT",
  "email": "chilumulamadhulika@klh.edu.in",
  "createdAt": "2026-09-21T07:04:09.524Z",
  "updatedAt": "2026-09-21T07:04:09.524Z",
  "__v": 0
}
```

PATCH → changed age from 23 to 25:

Chilumula Madhulika's Workspace

PATCH http://localhost:3000

http://localhost:3000/students/6ab0d7b74bd72bb2efd336e5

PATCH http://localhost:3000/students/6ab0d7b74bd72bb2efd336e5

raw JSON

```
{
  "age": 25
}
```

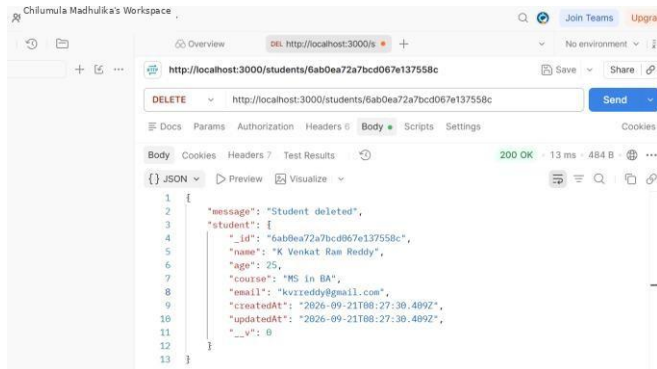
200 OK · 594 ms · 444 B

```
{
  "_id": "6ab0d7b74bd72bb2efd336e5",
  "name": "K Venkat Ram Reddy",
  "age": 25,
  "course": "MS in BA",
  "email": "kvrreddy@gmail.com",
  "createdAt": "2026-09-21T07:07:35.746Z",
  "updatedAt": "2026-09-21T08:24:56.224Z",
  "__v": 0
}
```

After modification of age output in mongosh:

```
{
  _id: ObjectId('6ab0ea72a7bcd067e137558c'),
  name: 'K Venkat Ram Reddy',
  age: 25,
  course: 'MS in BA',
  email: 'kvrreddy@gmail.com',
  createdAt: ISODate('2026-09-21T08:27:30.409Z'),
  updatedAt: ISODate('2026-09-21T08:27:30.409Z'),
  __v: 0
}
```

DELETE:



Observation of output in mongosh:

```
student_records> db.students.find()
[
  {
    _id: ObjectId('6ab0d6e94bd72bb2efd336e4'),
    name: Madhulika
    age: 18,
    course: 'CSIT',
    email: chilumulamadhulika@klh.edu.in
    createdAt: ISODate('2026-09-21T07:04:09.524Z'),
    updatedAt: ISODate('2026-09-21T07:04:09.524Z'),
    __v: 0
  }
]
student_records> |
```

Folder Structure:

Week_7_StudentRecordsAPI

- ├─ controllers
 - └─ studentController.js
- ├─ models
 - └─ Student.js
- ├─ routes
 - └─ studentRoutes.js
- ├─ server.js
- └─ package.json

Code: package.json

```
{
```

```

    "name": "week_7_studentrecordsapi",
    "version": "1.0.0",
    "type": "module",
    "description": "",
    "main": "server.js",
    "scripts": {
      "start": "node server.js",
      "dev": "nodemon server.js"
    },
    "keywords": [],
    "author": "",
    "license": "ISC",
    "dependencies": {
      "express": "^5.2.1",
      "mongoose": "^9.10.1"
    },
    "devDependencies": {
      "nodemon": "^3.1.14"
    }
  }
}

models/Student.js import
mongoose from 'mongoose';
const studentSchema = new
mongoose.Schema({  name: {    type:
String,      required: true,    trim: true
},  age: {    type: Number,    required:
true,    min: 16,    max: 100  },
course: {    type: String,    required:
true,    trim: true  },  email: {
type: String,
    required: true,
unique:      true,
lowercase: true
}  },
{
  timestamps: true
}); const Student = mongoose.model('Student',
studentSchema);

```

```
export default Student;
```

```

controllers/studentController.js import
Student from '../models/Student.js';
export const createStudent = async (req, res) => {
try {  const student = await
Student.create(req.body);
res.status(201).json(student);
} catch (error) {  res.status(400).json({
error: error.message });
}
}

```

```

}; export const getAllStudents = async (req, res)
=> { try { const students = await
Student.find(); res.status(200).json(students);
} catch (error) { res.status(500).json({
error: error.message });
}
};
export const getStudentById = async (req, res) => {
try {
const student = await Student.findById(req.params.id);

if (!student)
return res.status(404).json({ message: 'Student not found' });

res.status(200).json(student);
} catch (error) {
res.status(400).json({ error: error.message });
}
}; export const updateStudent = async (req, res)
=> { try {
const student = await Student.findByIdAndUpdate(
req.params.id, req.body,
{ new: true, runValidators: true }
); if (!student) return res.status(404).json({
message: 'Student not found' });

res.status(200).json(student);
} catch (error) { res.status(400).json({
error: error.message });
}
}; export const deleteStudent = async (req, res) => { try {
const student = await Student.findByIdAndDelete(req.params.id);
if (!student) return res.status(404).json({ message:
'Student not found' });
res.status(200).json({
message: 'Student deleted',
student
});
} catch (error) { res.status(400).json({
error: error.message });
}
};

```

routes/studentRoutes.js

```

import express from
'express';
import {
createStudent,
getAllStudents,
getStudentById,

```

```

updateStudent,
deleteStudent
} from '../controllers/studentController.js';
const router =
express.Router();
  router.post('/', createStudent);
router.get('/', getAllStudents);
router.get('/:id', getStudentById);
router.patch('/:id',
updateStudent);
router.delete('/:id',
deleteStudent);

```

```
export default router;
```

server.js

```

import express from 'express'; import mongoose from
'mongoose'; import studentRoutes from
'./routes/studentRoutes.js';
const app =
express(); const PORT
= 3000;
  app.use(express.json());
  app.get('/', (req, res) => {    res.json({
message: 'Student Records API is running'
  });
}); app.use('/students',
studentRoutes);
  mongoose.connect('mongodb://127.0.0.1:27017/student_records')
    .then(() => {
console.log('MongoDB connected');
      app.listen(PORT, () => {    console.log(`Server running
on http://localhost:${PORT}`);
      });
    })
    .catch((error) => {
      console.error('MongoDB connection failed:', error.message);
    });

```